

┌ An Integrated Framework for Planning Urban Autonomous Mobile Robot Paths ┐

University of Seoul

Ph.D. Program in Transportation Engineering

Presenter: Seongjin Park

01 Introduction

Research Background and Current Focus

- My research integrates autonomous robotics with urban and transportation planning, focusing on robot operations on sidewalks, transportation networks, and parking facilities.

Background & Research

Academic Background

- B.S. in Mechanical and Automotive Engineering
- M.S. in Smart City
- Ph.D. Student in Transportation Engineering
- Interdisciplinary Program in Urban Big Data Convergence

Current Research

- Autonomous Mobile Robots
- Urban Transportation Networks
- Path Planning and Simulation
- Urban Digital Twins
- Robot-Pedestrian Interaction
- Parking Robot Path Planning and Operations

Selected Publications

- Published and accepted research on autonomous mobile robots, path planning, urban networks, and robot-pedestrian interaction.

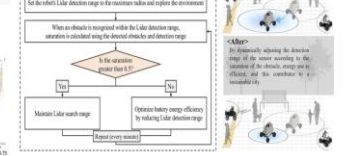
RESEARCH ARTICLE | FEBRUARY 16 2026
Network construction strategies for the activation of autonomous mobile robots
Seungjae Park, Sejin Min, Minja Choi, Seon Kim, Seungjae Lee
† Author & Article Information
Proceedings of the Institution of Civil Engineers - Municipal Engineer 1-11
https://doi.org/10.1680/jmuen.25.00118 Article history

Autonomous mobile robots (AMRs) are increasingly recognised as a sustainable solution to urban challenges such as environmental pollution and labour shortages. However, dynamic obstacles in urban areas with high pedestrian density hinder their safe and efficient operation, making it increasingly important to identify favourable segments and routes for them to navigate. This study proposes the construction of a robot-specific network that incorporates link-level resistance factors, including sidewalk width and obstacle locations, as weighted attributes. Within the study area, origin-destination pairs were defined, and urban network analysis was applied to identify high-activity pedestrian segments and derive potential paths. The applicability of the proposed network was validated through field experiments using AMRs under real-world pedestrian conditions. The results confirmed that segments with frequent pedestrian robot conflicts were associated with significant travel delays, whereas paths with fewer conflicts demonstrated superior travel performance. Overall, the study provides empirical evidence supporting the necessity of robot-specific networks that explicitly incorporate pedestrian density and congestion.

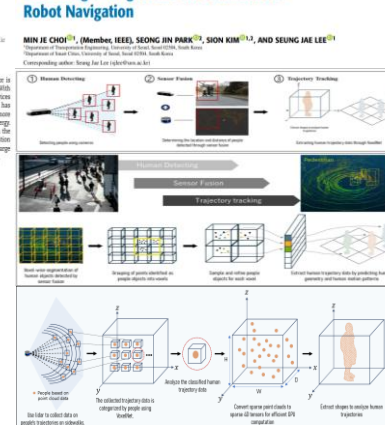


Energy efficient robot operations by adaptive control schemes
Minja Choi, Seungjae Park, Sejin Min, Seon Kim, Seungjae Lee
† Author & Article Information
Proceedings of the Institution of Civil Engineers - Municipal Engineer 1-11
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Energy efficiency is key to achieving the Sustainable Development Goals globally. Energy consumption in the transport sector is constantly increasing, and governments are implementing policies to reduce car use by shifting the focus from roads to walking. With the use of performance policies, autonomous mobile robots (AMRs) are becoming increasingly useful. Autonomous robot services are being used in various fields such as traffic management, logistics, and personal mobility assistance. However, AMR research has focused on technology development, route planning, and cost reduction, with relatively little research on how to make robots more energy efficient. As these autonomous robot services become more popular, there is a need to discuss how to efficiently use energy. This study analyses the characteristics of the hardware required for AMRs to operate. In particular, the density of obstacles in the surrounding environment is defined as a situation for the use of AMRs, and the effectiveness of the proposed policy optimisation reinforcement learning algorithm is analysed to propose an energy efficiency plan for the situation density in the future. A large number of robots are expected to be used, and efficient energy use of such hardware will contribute to building sustainable cities.



Human-Aware Trajectory Optimization for Enhancing D* Algorithm for Autonomous Robot Navigation
Minja Choi, Seungjae Park, Sejin Min, Seon Kim, Seungjae Lee
† Author & Article Information
Proceedings of the Institution of Civil Engineers - Municipal Engineer 1-11
https://doi.org/10.1680/jmuen.25.00118 Article history



Patents and Startup

- Founder & CEO
- Founded September 23, 2025

Human Robotics

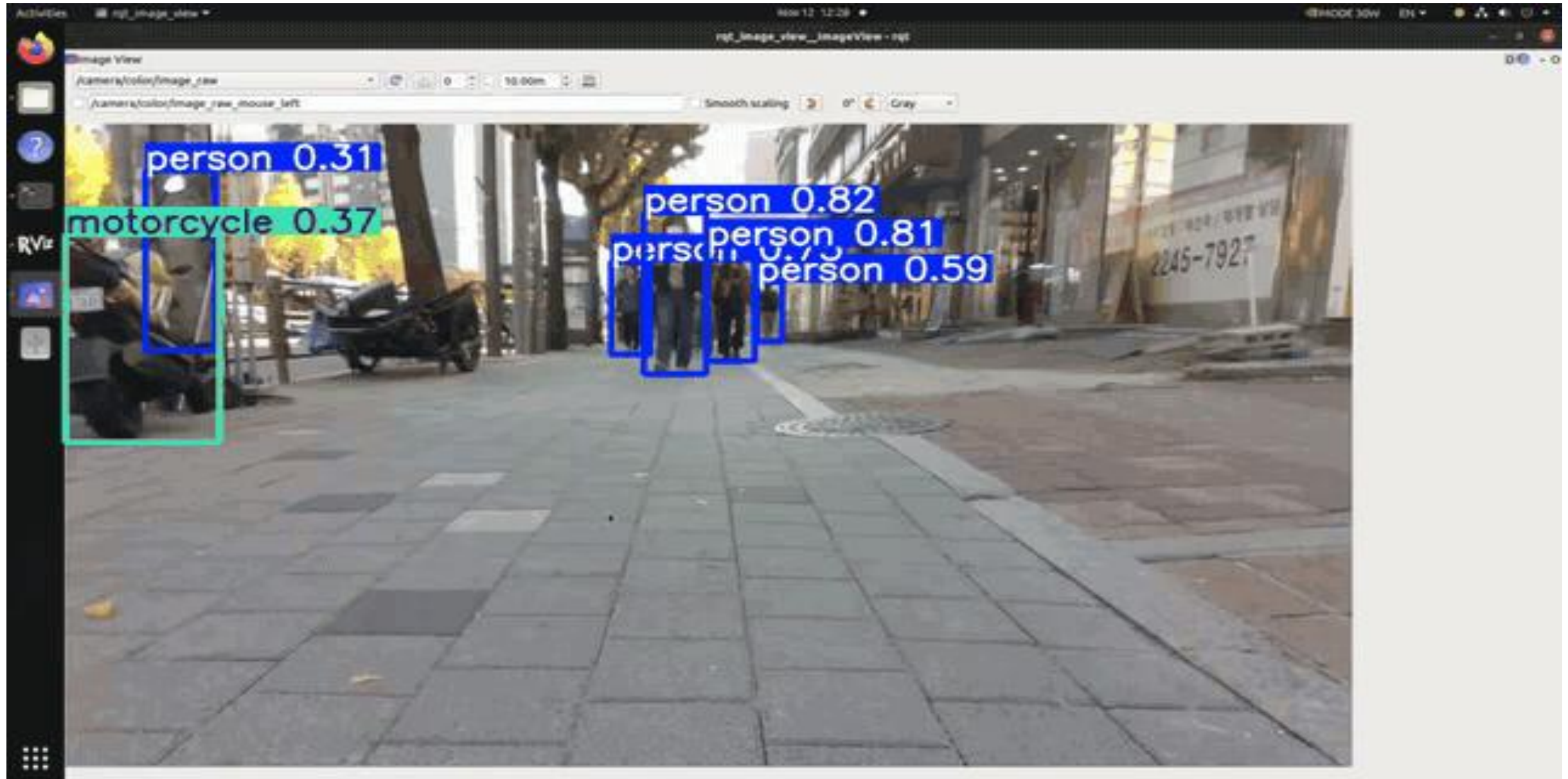
- 2 Registered Patents • 5 Patent Applications



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- 01. Introduction**
- 02. Literature Review**
- 03. Method**
- 04. Results**
- 05. Conclusion**

01 Introduction



✓ Origin of the Term Robot

- The concept of robots first emerged in literature before becoming a real-world technology.
- The term "robot" and Asimov's Three Laws shaped early ideas about how robots should work and interact with humans.



Isaac Asimov

- Born: January 2, 1920
- Died: April 6, 1992
- Occupation: Professor and Science Fiction writer
- Years active: 1939-1992
- Education: Ph.D. in Biochemistry, Columbia University

Isaac Asimov's Three Laws of Robotics

1. Protect Humans

- A robot must not harm a human being.

2. Obey Human Orders

- A robot must obey humans unless this conflicts with the First Law.

3. Protect Itself

- A robot must protect itself unless this conflicts with the first two laws

<Evolution of Robots>

Year	Contents
1921	• The term "robot" first appeared in Czech playwright Karel Capek's novel R.U.R.1921
1959	• The first industrial robot was developed by Joseph Engelberger and others at Unimate
1974	• Cincinnati Company developed T3, the first computer-controlled industrial robot
1979	• SCARA (Selective Compliance Assembly Robot Arm) was developed at Yamanashi University in Japan
1997	• Honda in Japan introduced P2, the first humanoid robot capable of climbing stairs (the predecessor to ASIMO)
1999	• Sony in Japan released AIBO (Artificial Intelligence Robot), the first pet robot
2003	• The mobile robot 'Spirit' conducted exploratory activities on Mars, operated by NASA in the United States
2006	• Boston Dynamics in the United States developed 'Big Dog', in collaboration with NASA and Harvard students



Source: <https://ko.wikipedia.org/wiki/%EB%A1%9C%EB%B4%87>

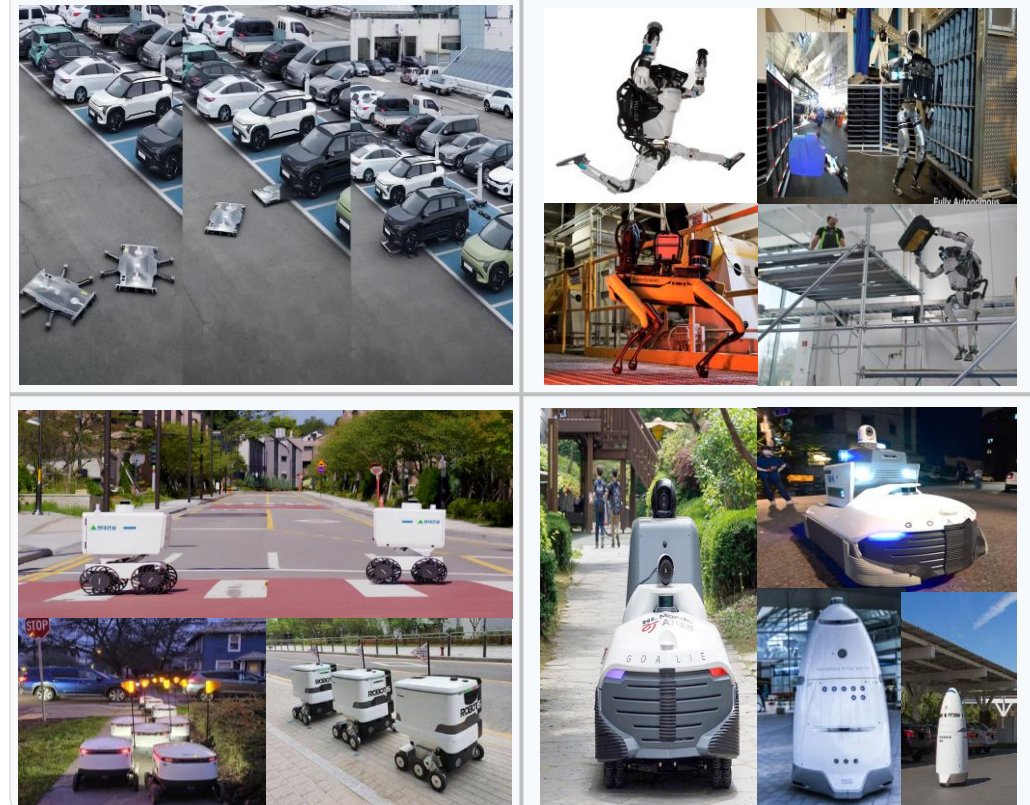
01 Introduction

✓ Robots Are Entering Urban Space

- My research focuses on how autonomous mobile robots (AMRs) and emerging mobility systems can be efficiently integrated into urban transportation networks.



Examples of Urban Mobile Robots

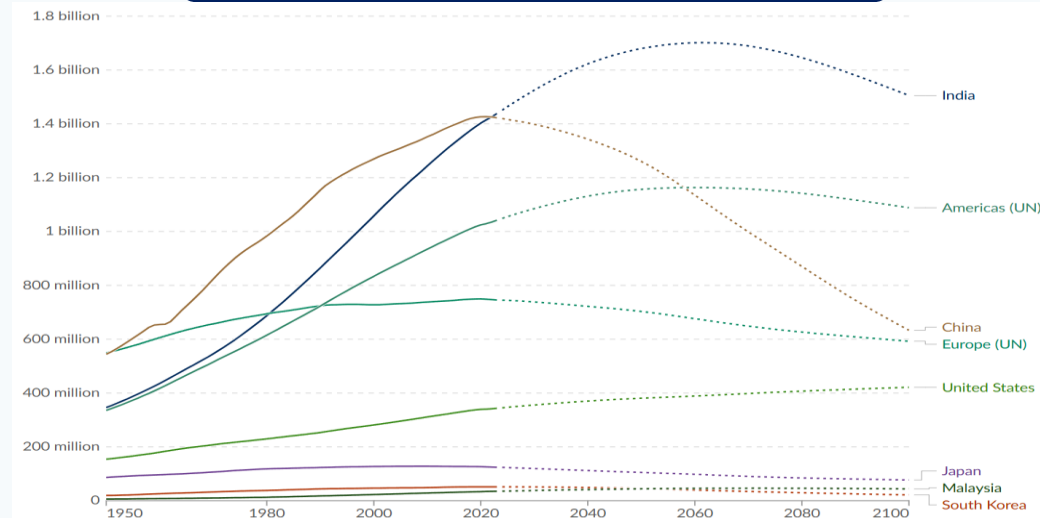


Source: Hyundai Motor Group, Reddit, Robotis, Hyundai Construction, HL Mando, The Electronic Times, Knightscope

Why Robots Are Needed: Population Decline and Labor Shortages

- Korea's population is projected to decline from 51.67 million in 2025 to 21.85 million in 2100.
- Declining working-age populations across OECD countries are increasing the need for automation and robotic support.

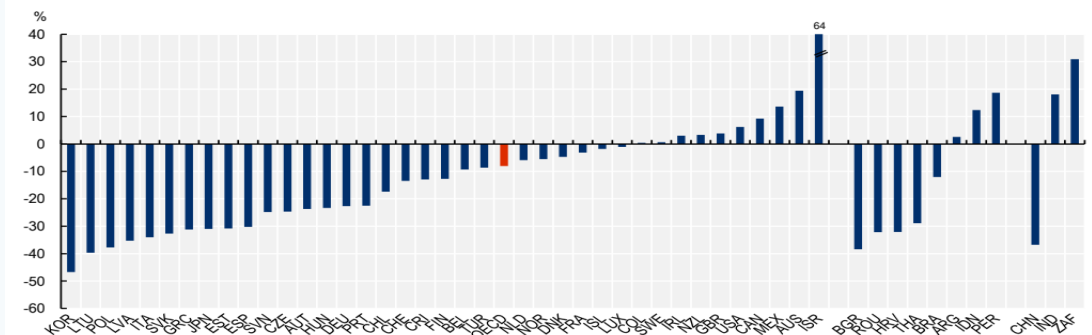
Population Trends, 1950-2100



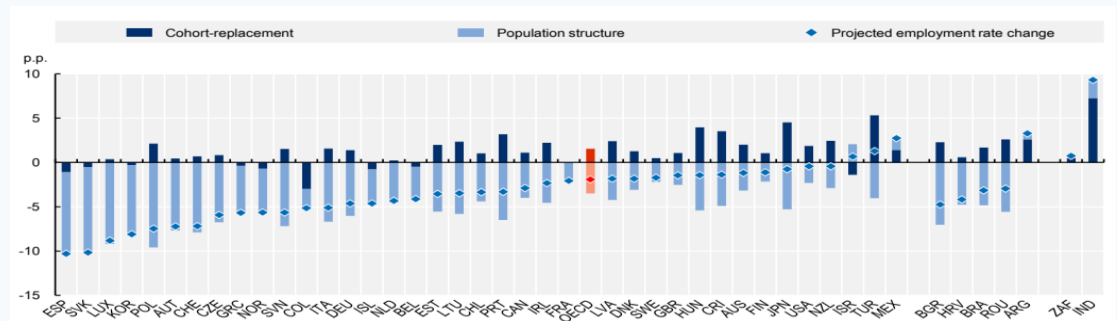
Population in 2025		Population in 2050		Population in 2100	
India	1.46 billion	India	1.68 billion	India	1.51 billion
China	1.42 billion	China	1.26 billion	Americas (UN)	1.09 billion
Americas (UN)	1.06 billion	Americas (UN)	1.16 billion	China	633.37 million
Europe (UN)	744.40 million	Europe (UN)	703.03 million	Europe (UN)	592.25 million
United States	347.28 million	United States	380.85 million	United States	421.28 million
Japan	123.10 million	Japan	105.12 million	Japan	76.85 million
South Korea	51.67 million	South Korea	45.14 million	Malaysia	44.03 million
Malaysia	35.98 million	Malaysia	44.29 million	South Korea	21.85 million

Declining Labor Supply and Population Aging

Decline in the Working-Age Population



Decline in the Employment-to-Population Ratio



source: Our World in Data, Population, 1950 to 2100

source: OECD Employment Outlook 2025

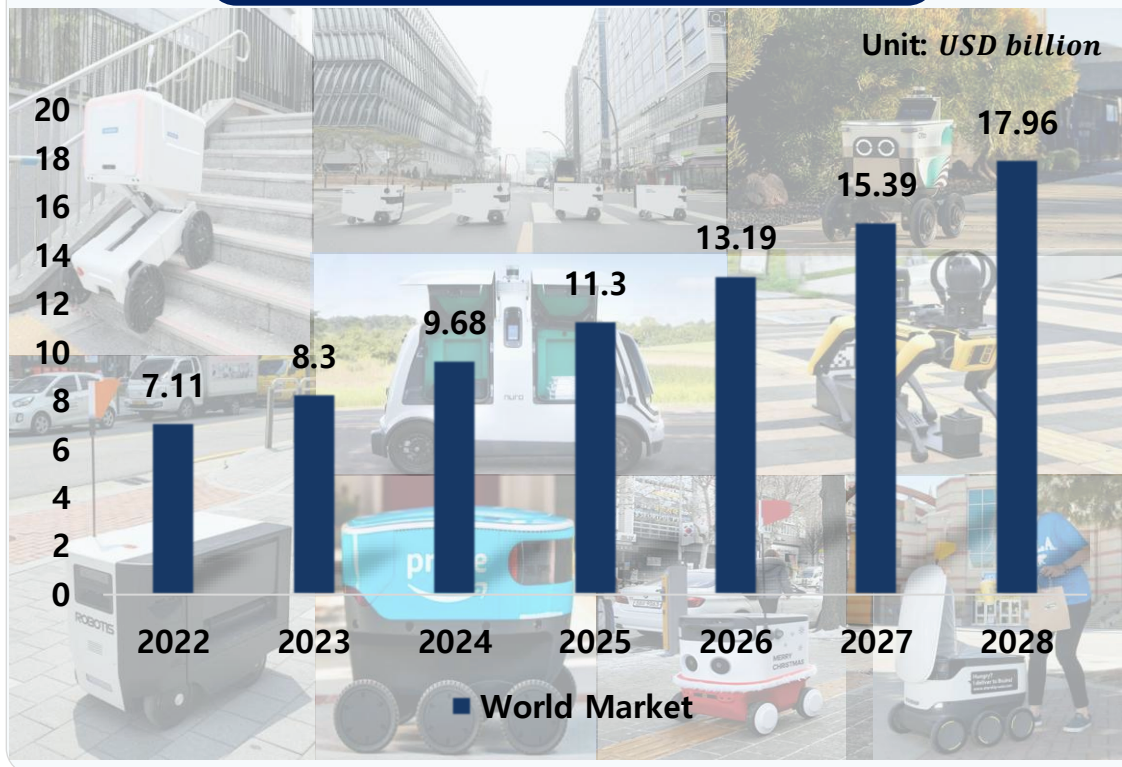
01 Introduction



✓ Growing Demand for Autonomous Mobile Robots (AMRs)

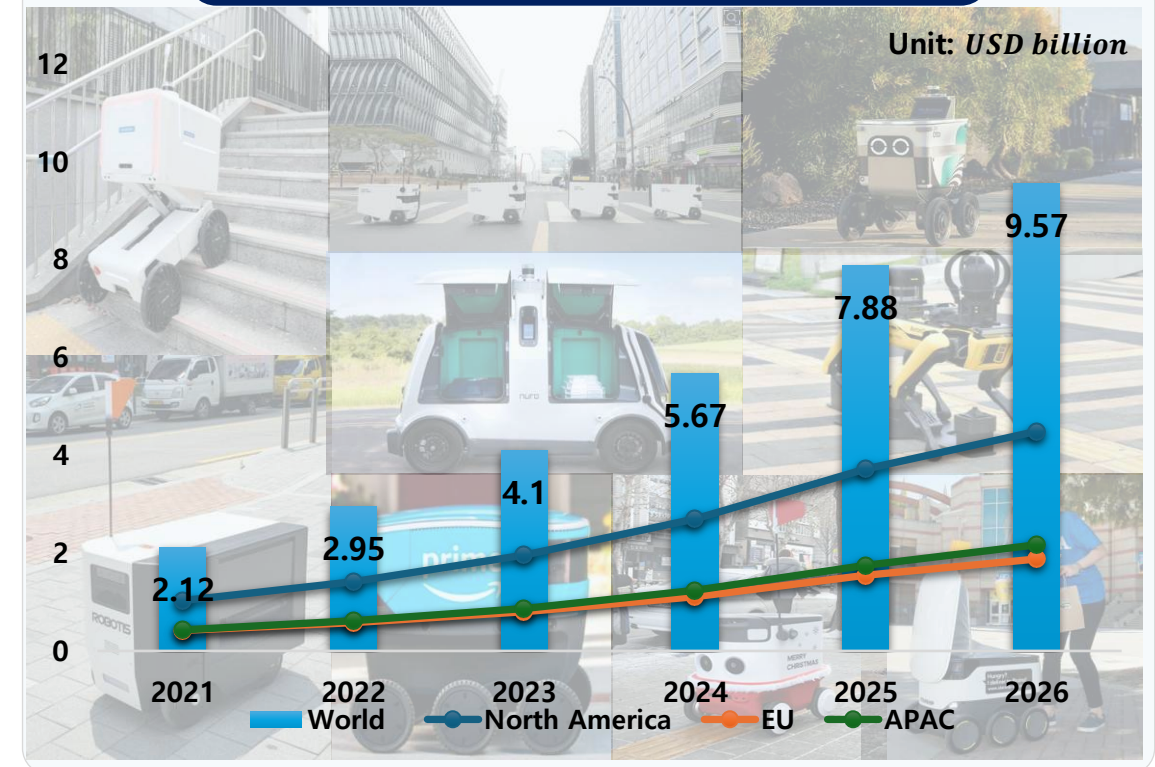
- The global AMR market is projected to grow from \$7.11 billion in 2022 to \$17.96 billion in 2028.
- This growth highlights the need to support efficient AMR operation while minimizing impacts on pedestrian safety and comfort.

Global AMR Market Size



source: Marketsandmarkets, Delivery Robot Market, 2021

Autonomous Robot Market Size



source: ROBOTIS, Nuro, Domino, Starship Technologies, Amazon Scout, Boston Dynamics, MOBINN

📌 Literature Review 1 – Autonomous Mobile & Delivery Robots(AMRs, ADRs) & Sidewalk Delivery Robots(SADRs)

- Recent studies have examined ADR and SADR technologies, applications, and regulations for automated last-mile delivery.
- ADRs can reduce delivery time and operating costs, particularly in dense urban environments.

Category	Research Title	Author / Year	Key Finding
Autonomous Mobile Robot(AMR) & ADR(Autonomous Delivery Robot(ADR) & SADR(Sidewalk Autonomous Delivery Robot)	Autonomous robot-driven deliveries: A review of recent developments and future directions	Srinivas, S., Ramachandiran, S., & Rajendran, S. / 2022 (Transportation research part E)	• A comprehensive review presenting the first synthesis of ADR-related technologies, design, regulations, and operational issues, outlining future research directions and policy challenges
	Study of Sidewalk Autonomous Delivery Robots(SADRs) and Their Potential Impacts on Freight Efficiency and Travel	Jennings, D., & Figliozzi, M. / 2019 (Transportation Research Record)	• The study verifies that SADRs effectively reduce delivery cost and time in urban environments, emphasizing the importance of regulation and the potential for commercialization
	Multi-goal Path Planning Autonomous System for Picking up and Delivery Tasks in Mobile Robotics	Hernandez, K., Bacca, B., & Posso, B. / 2017 (IEEE Latin America Transactions)	• The modified LKH algorithm, based on non-Euclidean distance and Hamiltonian paths, is applied to improve multi-goal Path planning and validated through real-world robot experiments
	Potentialities of Drones and Ground Autonomous Delivery Devices for Last-Mile Logistics	Lemardelé et al. / 2021 (Transportation research part E)	• A comparative analysis of ADRs and drones proposes optimal operating strategies by population density, confirming ADRs' economic advantage in high-density areas
	Autonomous last-mile delivery robots: A literature review	Alverhed et al. / 2024 (European Transport Research Review)	• The study reviews the operation, infrastructure, regulation, and social acceptance of last-mile delivery robots (ADRs) • It highlights their potential to reduce costs and carbon emissions, while noting limited real-world validation and institutional readiness

📌 Literature Review 2 – Urban Network Analysis(UNA) and Robotability

- UNA represents urban streets as node–link networks and evaluates their spatial structure, accessibility, and connectivity.
- Robotability considers how pedestrian flow and street conditions affect robot mobility.
- Together, these approaches support network-based AMR path planning.

Category	Research Title	Author / Year	Key Finding
Urban Network Analysis (UNA)	The network analysis of urban streets: A primal approach	Porta, S., Crucitti, P., & Latora, V. / 2006 (Environment and Planning B: planning and design)	• Proposed a primal model representing roads as node–link structures, quantifying urban structural complexity through centrality analysis.
	Introduction. Urban networks and network theory: the city as the connector of multiple networks	Pflieger, G., & Rozenblat, C. (2010) (Urban Studies)	• Analyzed interactions among multiple urban networks, emphasizing the importance of viewing the city as an interconnected system.
	Urban Network Analysis Toolbox	Sevtsuk & Mekonnen / 2012 (Env. & Planning B)	• Developed the Urban Network Analysis (UNA) tool, applying distance-based centrality metrics to pedestrian networks and establishing the basis for walkability assessment.
	Integration of Pedestrian Networks and Transit Accessibility Using UNA	Jeong et al. / 2023 (Journal of Urban Technology)	• Applied UNA to examine pedestrian–transit connectivity, illustrating its potential to identify interaction points between humans and robots.
Robotability	The Robotability Score: Enabling Harmonious Robot Navigation on Urban Streets	Franchi, Parreira, Bu, Wendy Ju / 2025	• Introduced the Robotability score metric, integrating weighted factors such as pedestrian density, crowd dynamics, and walking flow. • Evaluated robotability for New York City streets and validated results through real-world robot experiments.
	Group Surfing: A Pedestrian-Based Approach to Sidewalk Robot Navigation	Du, Hetherington, Oon et al. / 2021	• Proposed a “group-surfing” algorithm that lets robots follow pedestrian flow and avoid collisions in dense crowds.

02 Literature Review

Literature Review 3 – Pedestrian Infrastructure Mapping and Traversability-Aware Path Planning

- Classical algorithms such as Dijkstra and A* established the foundation for graph-based shortest-path planning.
- Advances in LiDAR and deep learning enabled pedestrian infrastructure to be mapped as navigable networks.
- Recent approaches integrate geometric and semantic information to support traversability-aware robot navigation.

Category	Research Title	Author / Year	Key Finding
Pedestrian Infrastructure Mapping and Traversability-Aware Path Planning	A Note on Two Problems in Connexion with Graphs	E. W. Dijkstra / 1959	• Introduced a systematic algorithm for finding the shortest path between nodes in a weighted graph. • Established the theoretical foundation for graph-based global path planning
	A Formal Basis for the Heuristic Determination of Minimum Cost Paths	P. E. Hart, N. J. Nilsson, B. Raphael / 1968	• Proposed the A* algorithm, which uses heuristic information to search for minimum-cost paths efficiently. • Provided the basis for optimal and computationally efficient robot path planning.
	Morphological Operations to Extract Urban Curbs in 3D MLS Point Clouds	B. Rodríguez-Cuenca, S. García-Cortés, C. Ordóñez, M. C. Alonso / 2016	• Developed an automated method for extracting curbs and road boundaries from mobile laser-scanning point clouds. • Demonstrated how physical pedestrian infrastructure can be converted into navigational constraints.
	A network-level sidewalk inventory method using mobile LiDAR and deep learning	Q. Hou, C. Ai / 2020	• Combined mobile LiDAR, deep learning, and geometric analysis to construct a network-level sidewalk inventory. • Extracted sidewalk width, cross slope, and longitudinal slope for accessibility and Path-planning applications.
	ViPlanner: Visual Semantic Imperative Learning for Local Navigation	P. Roth, J. Nubert, F. Yang, M. Mittal, M. Hutter / 2024	• Introduced a learning-based navigation framework that incorporates geometric and semantic information into traversability assessment. • Enabled zero-shot navigation in complex real-world environments using traversability-aware cost maps.

03 Method – Research Framework



1. Robot-Specific Network Construction

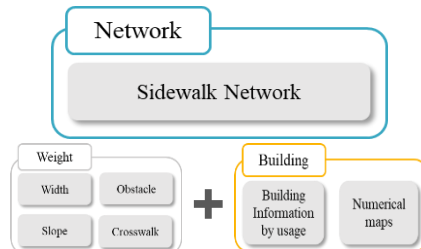
- Construct a robot-specific pedestrian network by integrating sidewalk and urban-environment data.

Autonomous Robot Platform



Study Area: University of Seoul Campus and Jeonnon-dong

Sidewalk Network and Attribute Integration



2. Urban Network Analysis (UNA)

- Analyze network-based pedestrian movement potential and select representative paths using Betweenness



$$\text{Betweenness}[k]^r = \sum_{i \neq j \neq k \in d[i,j] \leq r_i} \frac{N_{d[i,j]}[k]}{N_{d[i,j]}} * W[i,j] * W[k]$$

3. Robot Simulation

- Evaluate path-specific robot performance through 100 autonomous-navigation simulations

Simulation inputs
 $p \in \{A, B, C\}, N_p \approx 160 - 170$



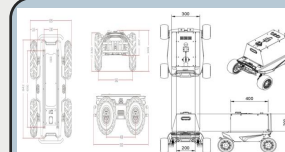
Performance Measure

$$T_{p,r} = t_{p,r}^{\text{arrival}} - t_{p,r}^{\text{departure}}$$

$$\bar{T}_p = \frac{1}{100} \sum_{r=1}^{100} T_{p,r}$$

4. Field Testing with SCOUT 2.0

- Conduct field tests along the same paths using the SCOUT 2.0 robot



Field-Test Setup

$$p \in \{A, B, C\}$$

$$r = 1, 2, \dots, 8$$

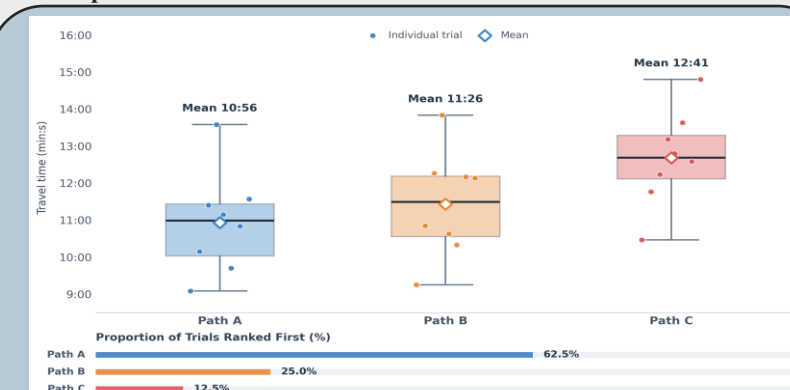
Performance Measure

$$T_{p,r} = t_{p,r}^{\text{arrival}} - t_{p,r}^{\text{departure}}$$

$$\bar{T}_p = \frac{1}{8} \sum_{r=1}^8 T_{p,r}$$

5. Integrated Results Comparison

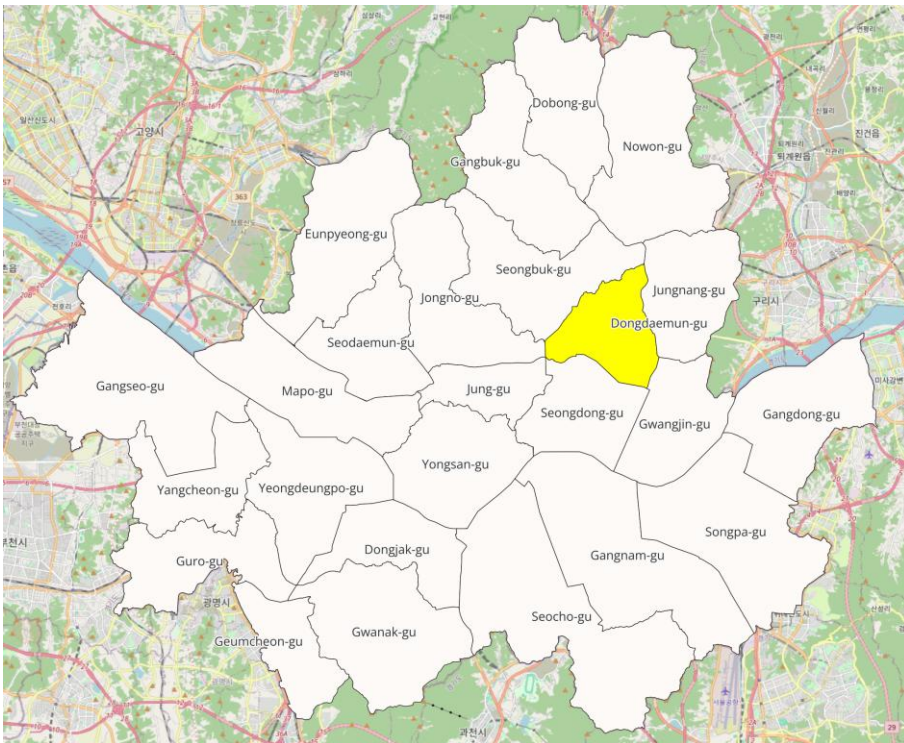
- Compare the UNA, simulation, and field-test results across the selected paths



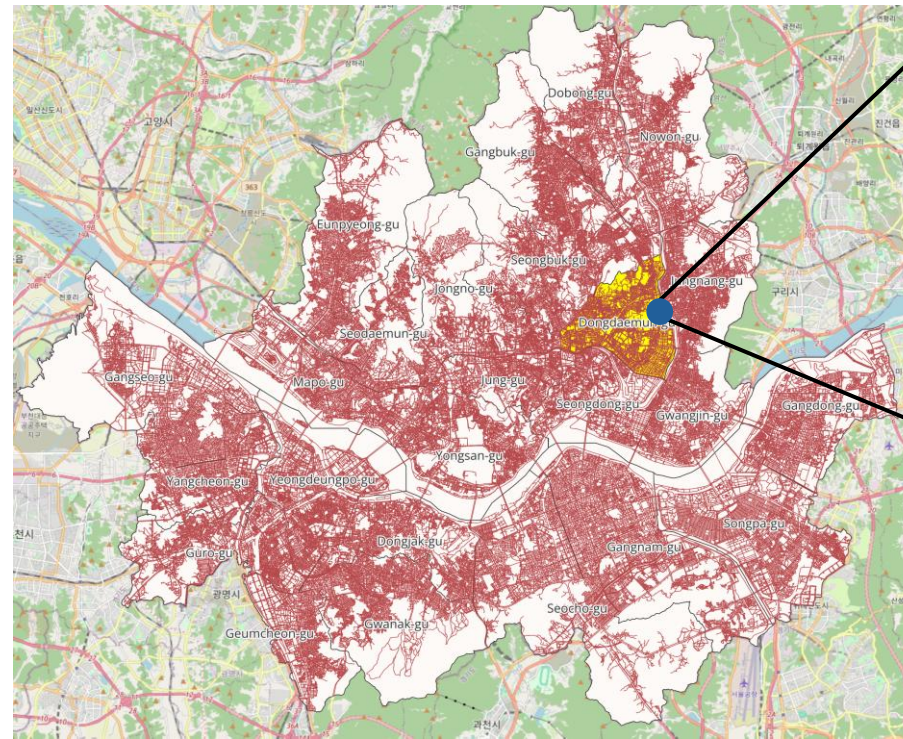
✓ Spatial Scope and Pedestrian Network

- The spatial scope was narrowed from Seoul to Dongdaemun-gu and Jeonnong-dong using the citywide pedestrian network.
- The simulation and field-test paths were defined around the University of Seoul campus based on the robot's practical operating range

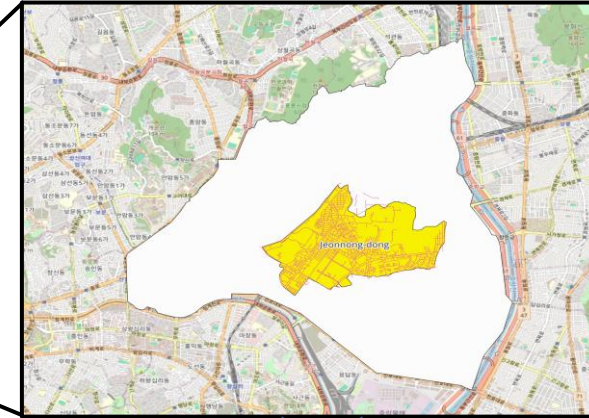
Administrative Districts of Seoul



Seoul Pedestrian Network



Dongdaemun-gu and Jeonnong-dong Study Area



Pedestrian Network of the Test Area



✓ Construction of a Robot-Specific Network

- Seoul Metropolitan Government's sidewalk network, digital map, and building-use data were integrated.
- The digital map supplemented missing sidewalk widths and building geometry, while building-use data were used to define O-D points for UNA.
- Sidewalk width, slope, obstacles, crosswalks, and bus stops were incorporated as link-specific attributes.

Data Utilization

Seoul City Sidewalk Network

- Outdoor mobile robots operate on pedestrian path-based navigation
- Stairs and impassable links removed

National Geographic Information

Platform Digital Map

- Robot navigation path width measurement
- Missing buildings added

Seoul City Building Information by Use

- Origin (Restaurants, Supermarkets, etc.)
- Destination (Apartments, Villas, Schools, etc.)

Robot Network Design Considerations



➤ Width (minimum 0.8m or more, bus stop)

➤ Intersection, curb



➤ Obstacles (bollards, signs, etc.)



➤ Parked vehicles

Network Link-Specific Weights

Category	Details
Network Width (Unit: m)	Calculating width per link in robot network (determining drivability)
Slope (Unit: %)	Using SMAP to calculate slope (determining drivability)
Obstacles (Unit: Count)	Estimating number of obstacles (using street view, direct survey)
Crosswalk (Unit: m)	Width and number calculation (outdoor mobile robot operation safety standards)
Bus Stops (Unit: Count)	Calculating the number of stops (for passenger safety during boarding and alighting)

➤ Designing a robot-dedicated network using link-specific weights

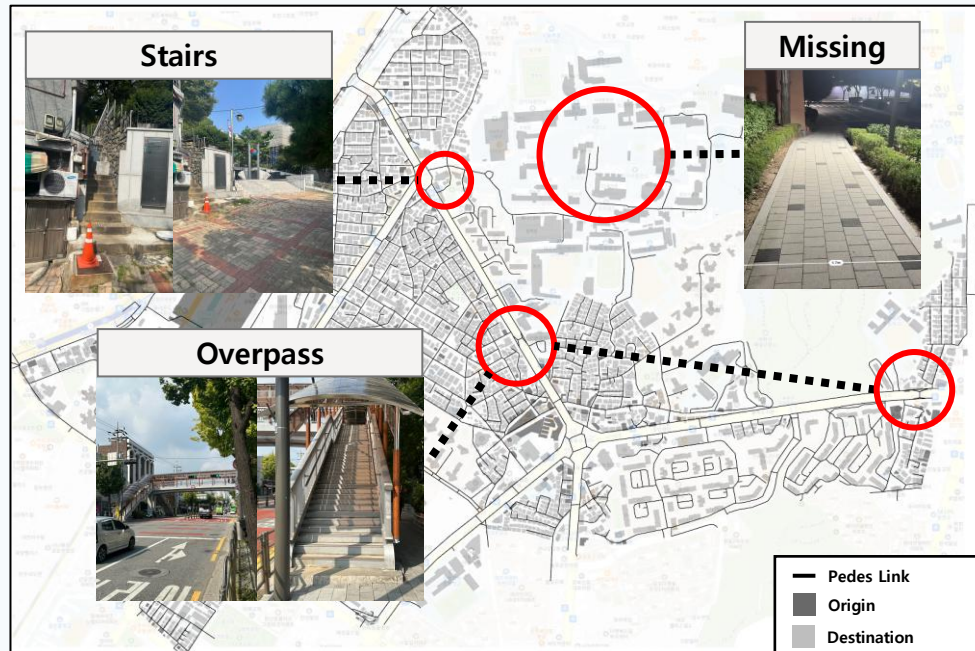
Constructing a **robot-specific network** to identify and evaluate candidate **AMR paths** using **UNA**.

✓ From the Sidewalk Network to a Robot-Specific Network

- Inaccessible links were removed, missing links were added, and path-relevant attributes and weights were assigned to each link.

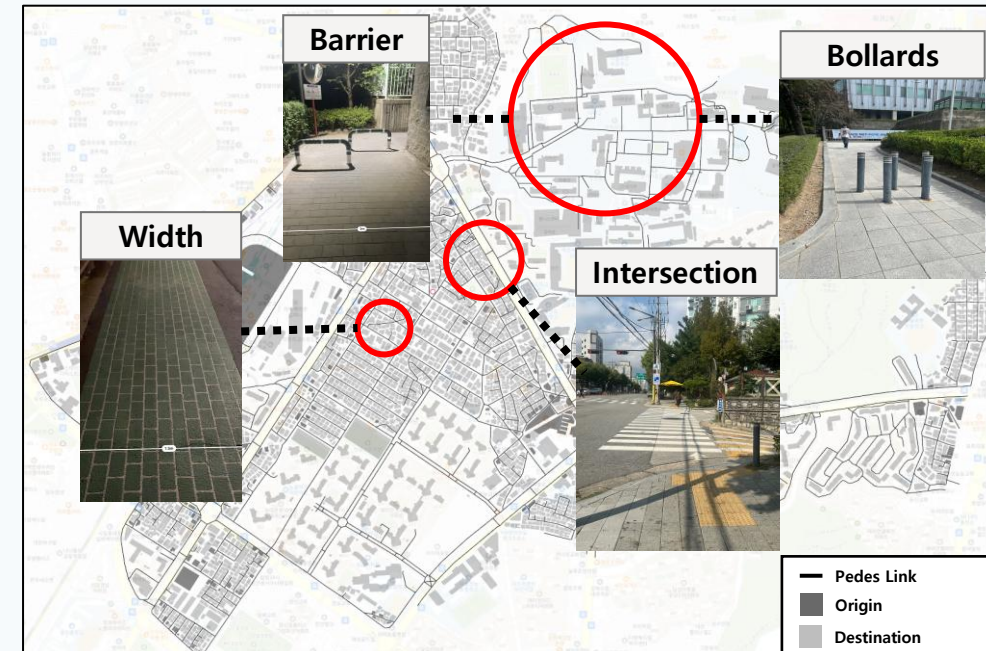
- Utilizes Seoul's sidewalk network and building data categorized by use
- Removes links (stairs, overpasses, etc.) inaccessible to delivery robots

Existing Sidewalk Network



- Adds links not reflected in the experimental area (Jeonnon-dong + UOS)
- Network by incorporating weighting factors (width, number of obstacles etc.)

Robot-Specific Network



✓ UNA Betweenness Analysis for Path Selection

- Betweenness measures how frequently a link appears on the shortest paths between origin–destination pairs.
- The weighted robot-specific network was used to identify important links and potential movement concentrations.
- In this study, Link Betweenness was used to identify links with greater potential for pedestrian movement.

➤ Betweenness can be calculated for nodes or links. This study uses Link Betweenness.

Betweenness equation

$$\text{Betweenness}[k]^r = \sum_{i \neq j \neq k \in d[i,j] \leq r_i} \frac{N_{d[i,j]}[k]}{N_{d[i,j]}} * W[i,j] * W[k]$$

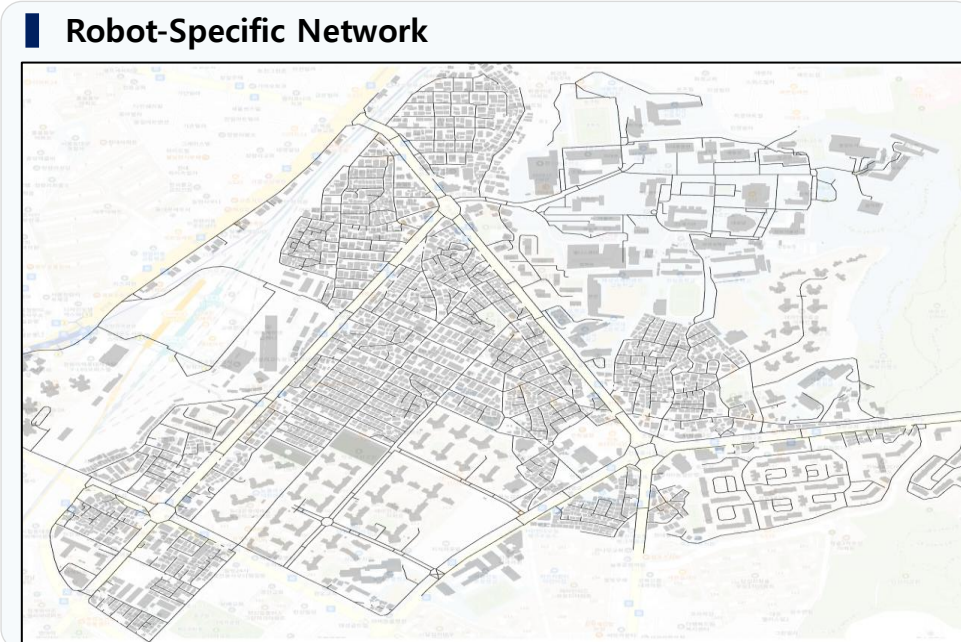
- i = Origin (ex: restaurant, grocery store)
- j = Destination (ex: residence, school)
- k = observation point
- r = spatial range/radius(e.g.: 1km)
- d[i, j] = shortest network path between origin and destination
- W[i] = weight of the origin
- W[j] = weight of the destination
- W[k] = weight of the observation point



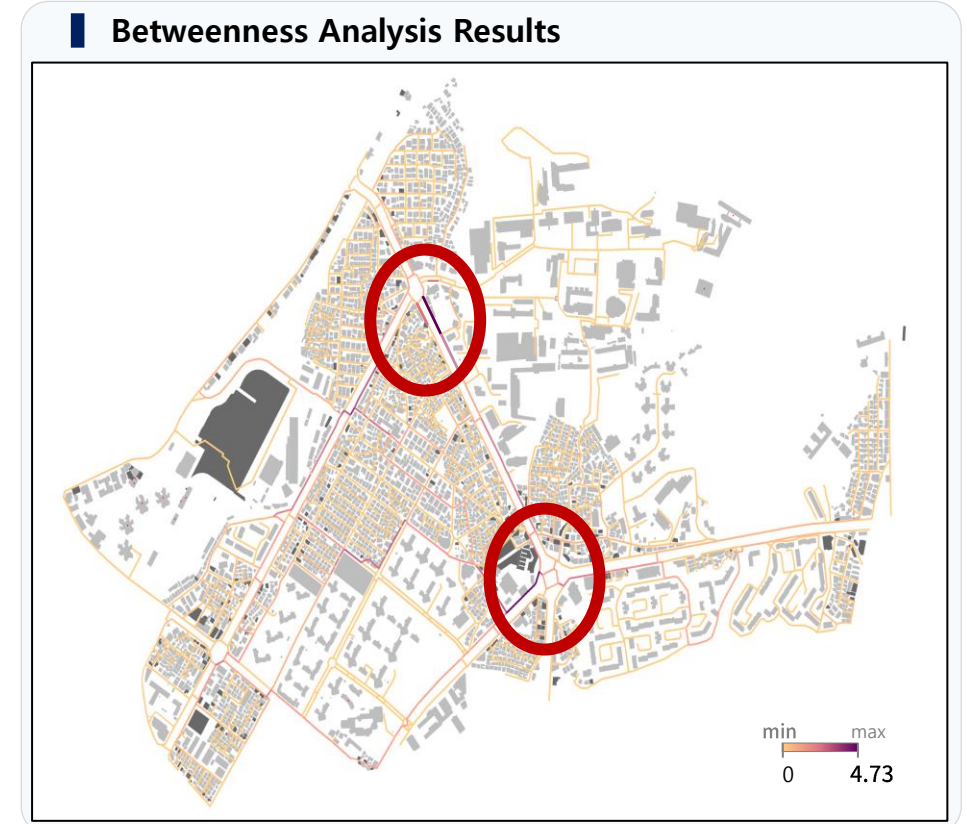
Source: Sevtsuk, A., & Mekonnen, M. (2012). Urban network analysis. *Revue internationale de géomatique-n*, 287, 305.

① UNA Betweenness Results and Test-Site Selection

- Weighted UNA Betweenness analysis was conducted to estimate network-based pedestrian movement potential within the robot-specific network.
- Links with darker colors indicate higher network-based pedestrian movement potential, with a maximum Betweenness value of 4.73.
- Two high-Betweenness areas were identified, and one was selected as the detailed test site for simulation and field testing.

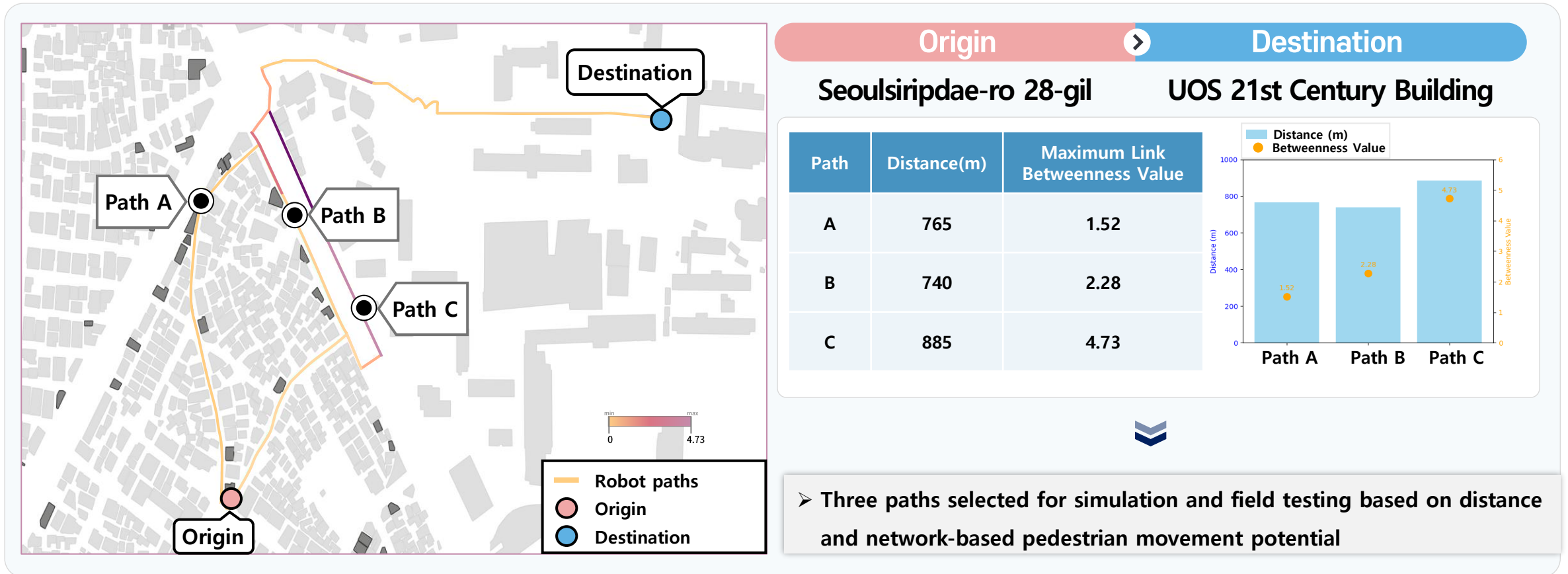


O/D	Land use classification	Number
Origin	Retail and Commercial Facilities(Restaurants, Supermarkets, etc)	232
Destination	Educational, Residential, and Healthcare Facilities(Schools, Apartments, etc)	3,601



✓ O-D and Alternative Path Selection Based on Betweenness

- Three alternative paths A, B, and C were defined between the same origin and destination.
- Path A had the lowest maximum Link Betweenness value 1.52, while Path C had the highest 4.73.
- The three paths were selected to represent different levels of network-based pedestrian movement potential.

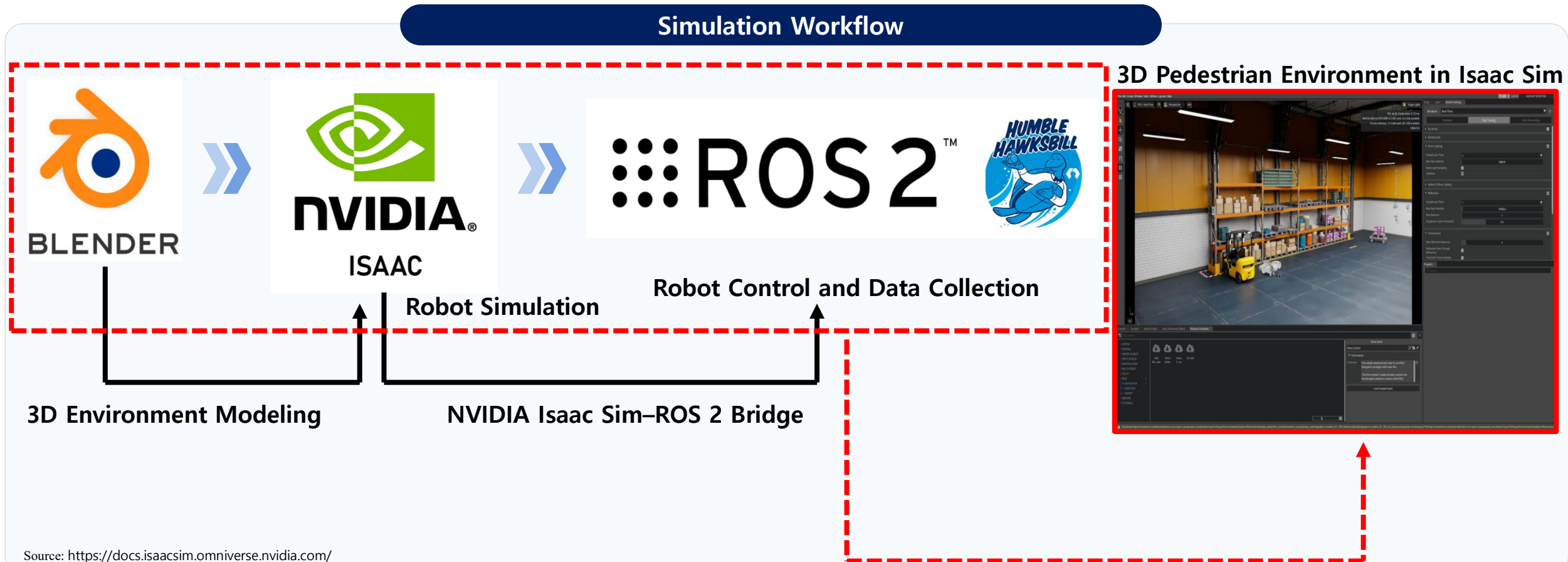


03 Method



✓ Simulation Environment Setup

- NVIDIA Isaac Sim is a robotics simulation platform for developing, testing, and evaluating AI-driven robots in physically based virtual environments.
- The 3D pedestrian environment developed in Blender was imported into NVIDIA Isaac Sim and integrated with ROS 2 for robot control and data collection.

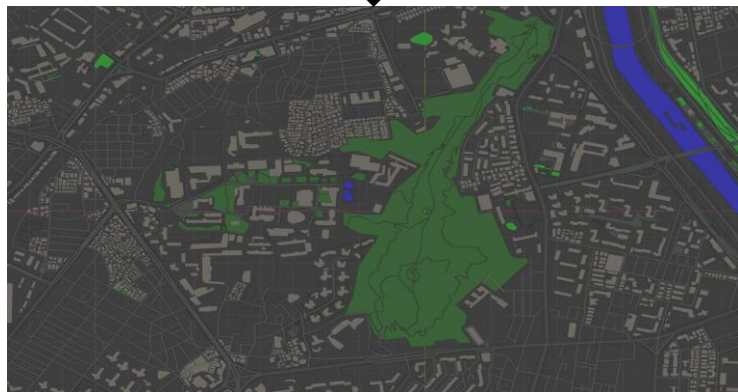


03 Method

✓ Blender-based 3D Map and Layer Classification

- Buildings, roads, sidewalks, and stairs were classified into separate objects and layers.
- Linear pedestrian layers were selected as candidate AMR paths, while stairs and level changes were defined as navigation constraints.

Blosm-based Map Data



Blender-based 3D Map



Layer Classification



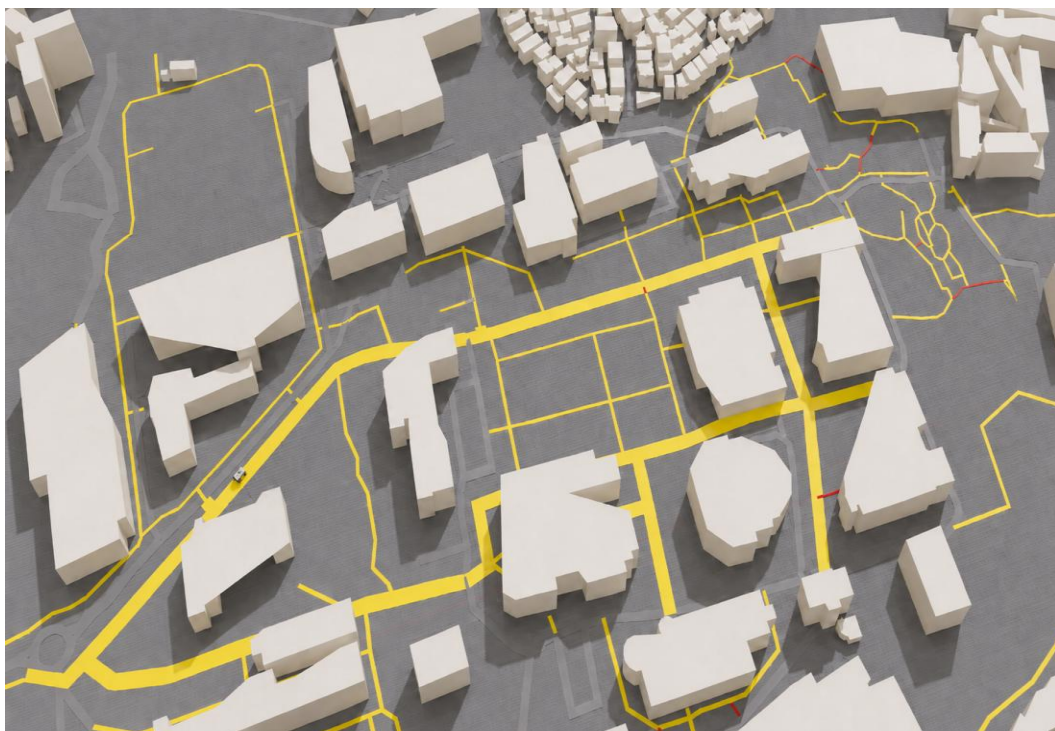
Color	Meaning	Applied Layer	Path Usage
Yellow	Linear Pedestrian path	Paths_Footway, Road_pedestrian	Candidate AMR Path
Gray	Areal pedestrian space	Areas_footway	Reference layer
Dark Gray / Black	Areal pedestrian zone	Areas_pedestrian	Reference layer
Gray	Roadway	Road	Excluded
Beige	Building	Building	Non-traversable
Red	Stairs	Steps, Stairs	Non-traversable

03 Method

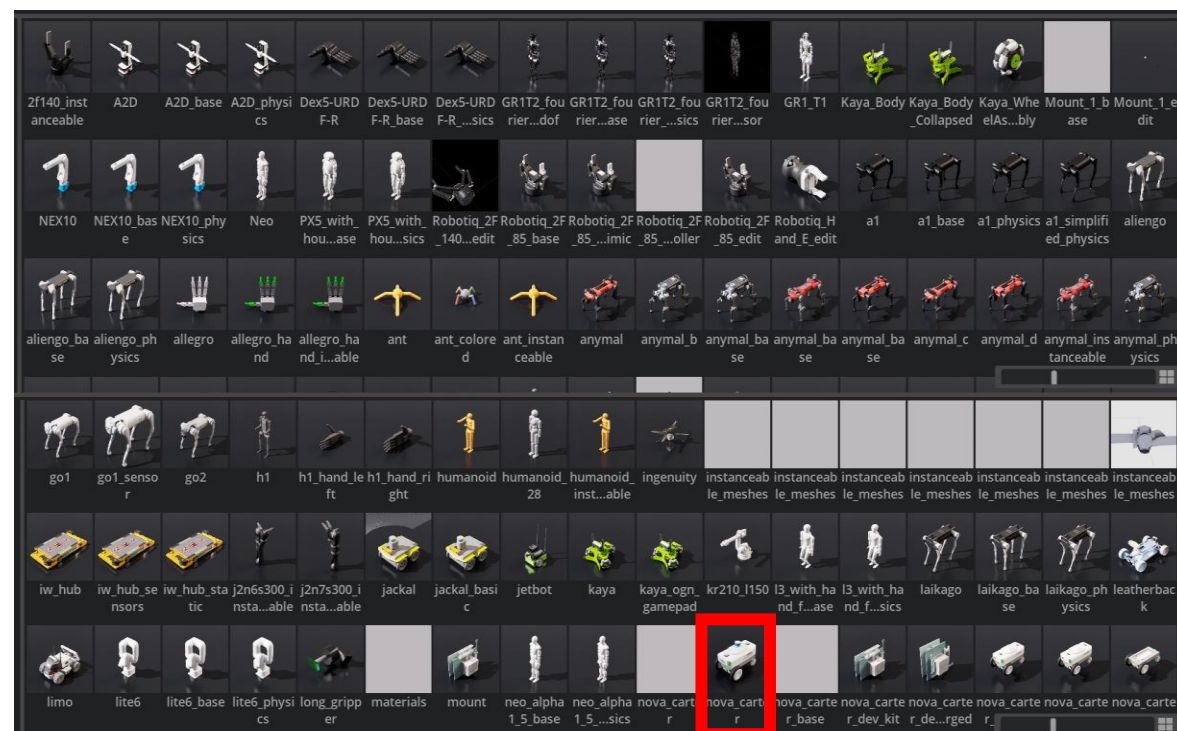
✓ AMR Simulation Setup

- The layer-based 3D pedestrian environment was imported into NVIDIA Isaac Sim as the simulation testbed.
- Nova Carter was selected as the simulation robot based on its camera, IMU, and LiDAR configuration.
- The robots were configured with similar footprints and the same maximum speed for comparison between the simulation and field test.

Layer-based 3D Testbed in Isaac Sim



Nova Carter Selection and Configuration



03 Method



✓ AMR Simulation Setup

- NVIDIA Isaac Sim was integrated with ROS 2 Humble through the ROS 2 Bridge.
- Control commands, driving states, and sensor data were transmitted through ROS 2 topics.
- ROS 2 data and simulation logs were used to derive robot trajectories, travel time, active pedestrian counts, and obstacle-avoidance events.



Topic	Data Type	Usage
• /chassis/imu	IMU Data	• Robot orientation and driving stability
• /chassis/odom	Position, Orientation, and Velocity	• Trajectory, travel distance, and average speed
• /cmd_vel	Velocity Command	• Linear and angular velocity commands
• /front_3d_lidar/lidar_points	3d Lidar Point Cloud	• Obstacle and environment perception
• /front_stereo_camera/left/image_raw	Camera Image	• Visual monitoring of the driving environment
• /tf	Coordinate Transform	• Robot and sensor frame relationships
• /clock	Simulation Time	• Simulation time synchronization

```

publishing #28: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.2, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.0))
publishing #29: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.2, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.0))
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    ..

```

03 Method

✓ Example of Simulation Scenario Setup

- Example setup showing how origin–destination points and pedestrian agents were configured in Isaac Sim.
- The actual experiments were conducted using Path-specific distances and approximately 160–180 pedestrian agents per Path.

Example: Path and O–D Setting



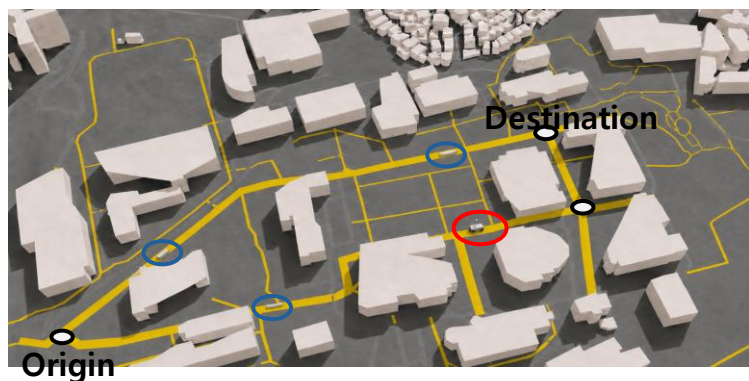
Example: 5 Pedestrians



Example: 15 Pedestrians



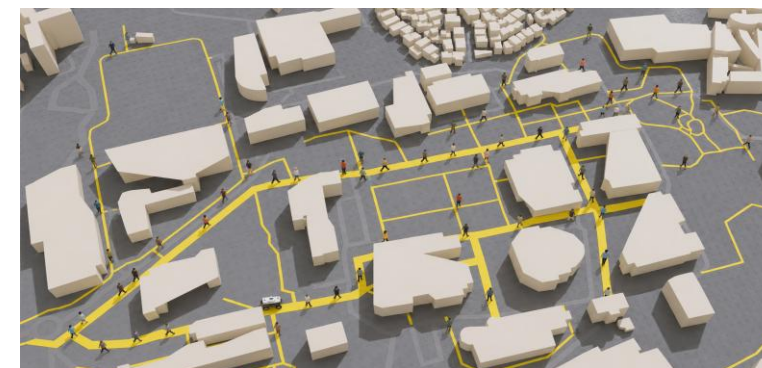
Example: Path and O–D Setting



Example: 10 Pedestrians



Example: 50 Pedestrians

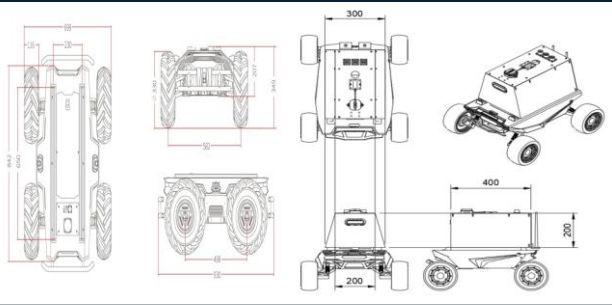


03 Method

✓ Field Test Setup and GPS Trajectory Mapping

- The field test used a SCOUT 2.0 outdoor AMR equipped with LiDAR, RTK-GPS, an IMU, and depth cameras.
- The maximum speed was set to 1.5 m/s to match the simulation and support safe operation in pedestrian environments.
- Latitude and longitude data collected through ROS 2 topics were mapped to analyze the robot's travel trajectory.

Robot Hardware Design and Customization



Sensors Mounted on the Robot



SCOUT 2.0 (AMR)



- Size: 930 x 699 x 349
- Wheelbase: 498 mm
- Max Climbing angle: 30°
- Max Speed: 1.5 m/s
- Weight: 68kg
- Max payload: 50kg

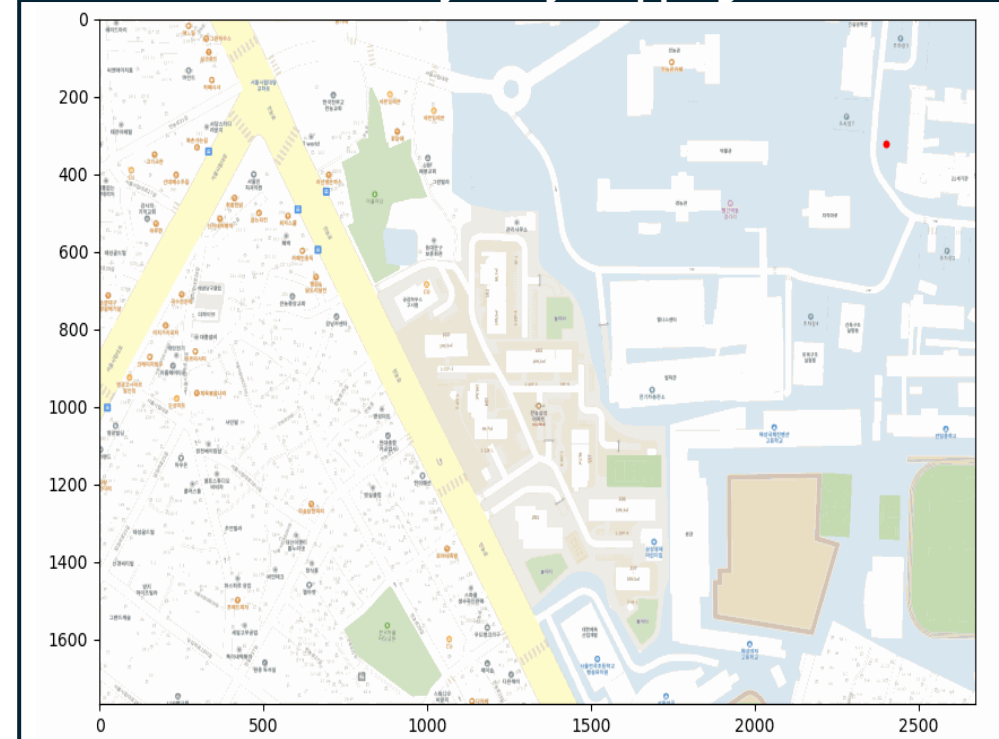


On-Campus Driving Test



- ❖ The maximum speed was set to 1.5 m/s to match the simulation and support safe operation in pedestrian environments.

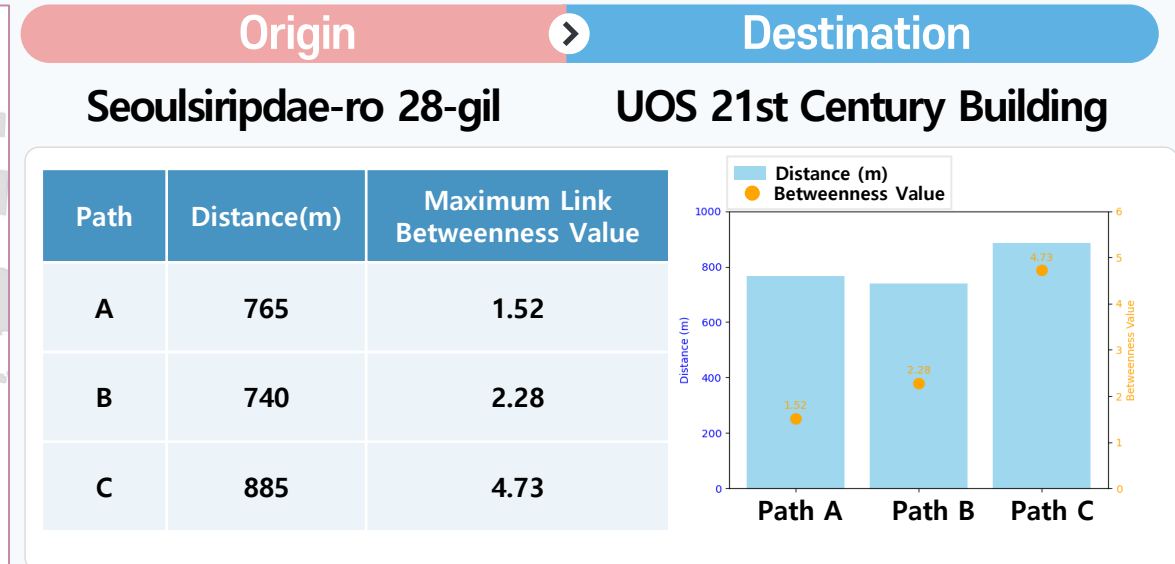
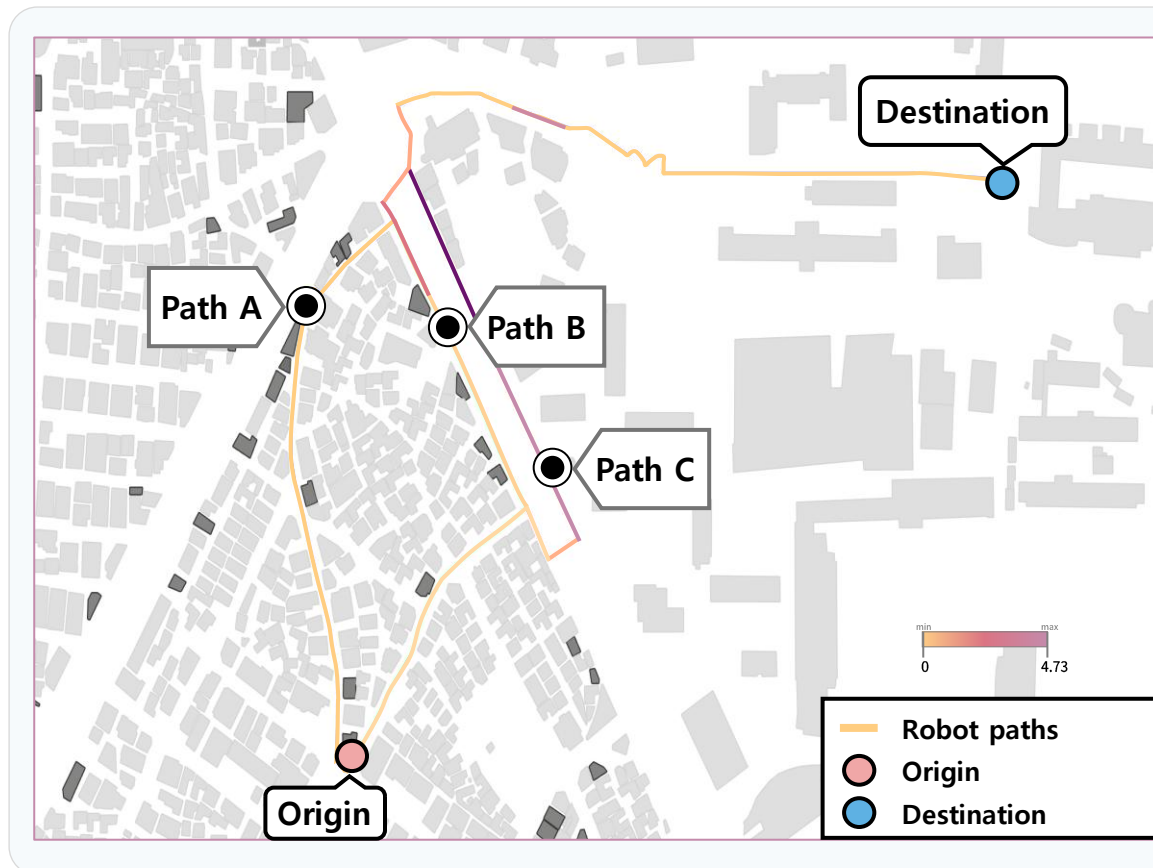
GPS Trajectory Mapping



04 Results

Expected Path Characteristics Based on UNA

- Path C was expected to present the most challenging operating conditions because it combined the longest distance with the highest maximum Link Betweenness.

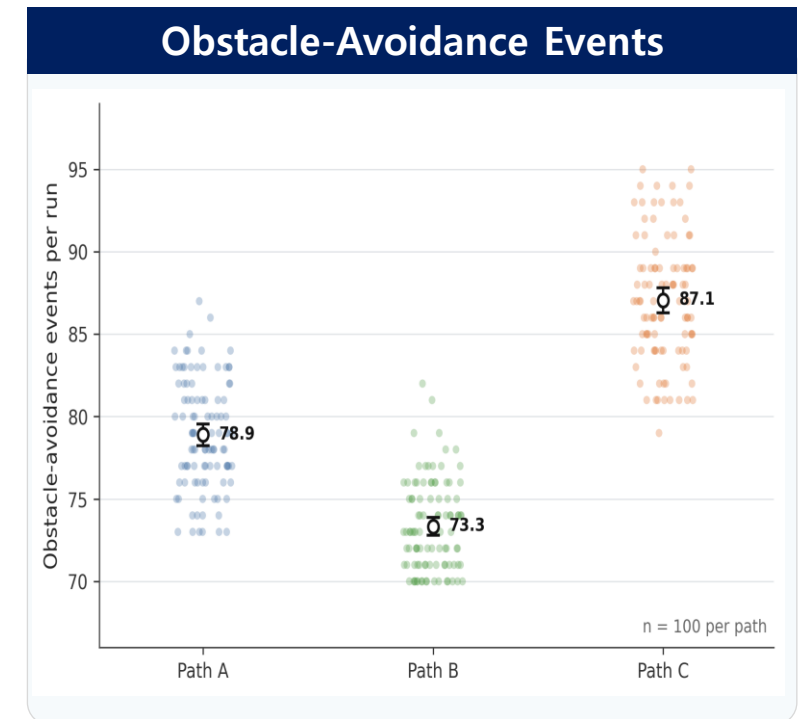
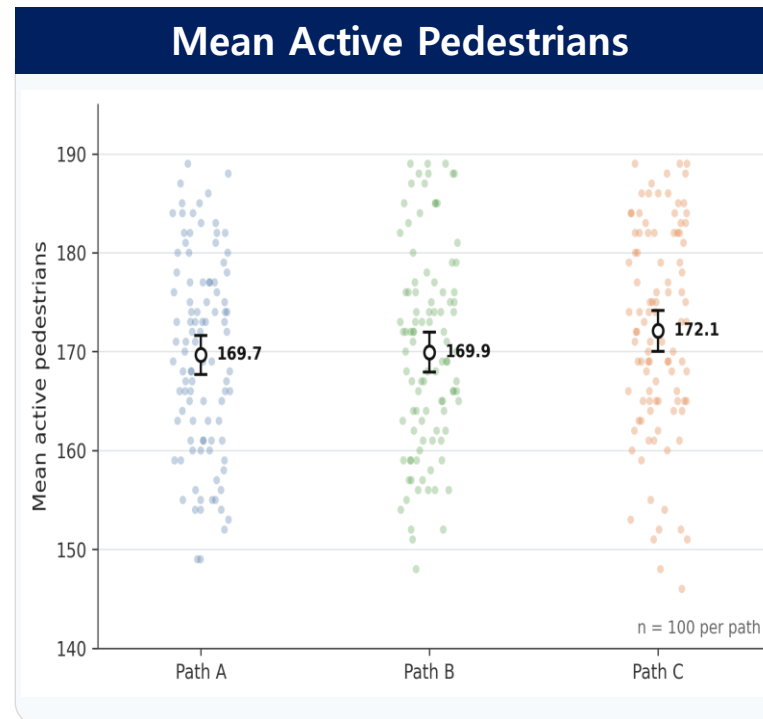
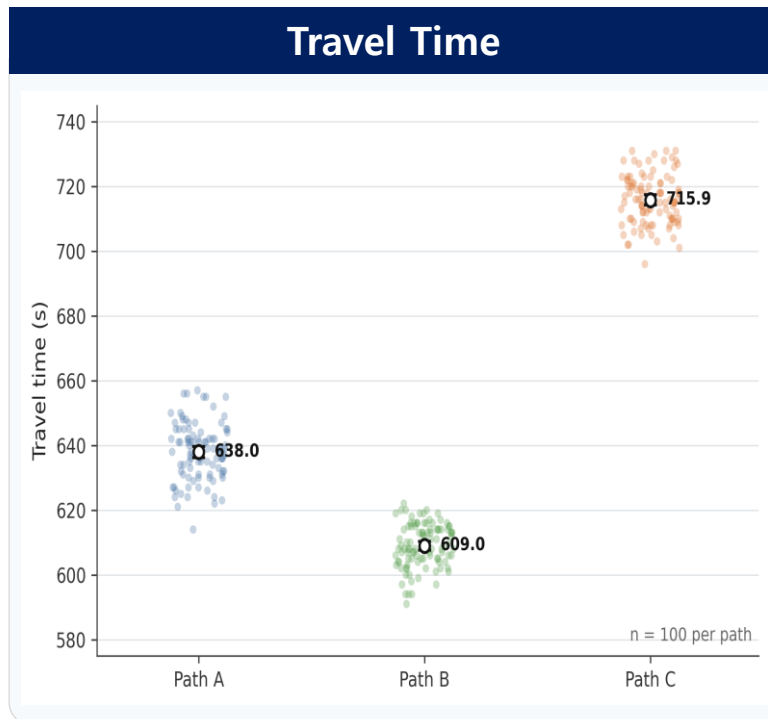


➤ Path C was expected to show the greatest travel delay due to its longer distance and higher network-based pedestrian movement potential.

04 Results

✓ Path-Level Performance in Isaac Sim

- Path C recorded the longest travel time and the highest Obstacle-Avoidance Events.
- Mean active pedestrians remained similar across the three paths.
- Under the current simulation conditions, travel-time differences appeared to be more closely associated with path distance and path-specific geometry.



04 Results

✓ Robot-Pedestrian Interactions Observed during Field Testing

- Pedestrian congestion caused travel delays at intersections, bus stops, and narrow sidewalk sections.
- At bus stops, boarding and alighting passengers temporarily obstructed the robot's path.
- In crowded areas, avoidance maneuvers reduced the robot's travel speed.

Bus Stop Area



Heavy Pedestrian Traffic



04 Results

☑ Infrastructure-Related Causes of Travel Delay

- At intersections, signal waiting and pedestrian crossings caused the robot to stop or slow down.
- At bus stops, interactions with boarding and alighting passengers temporarily interrupted robot movement.
- These observations demonstrate how real-world interaction conditions can affect robot travel performance.

Signalized Intersection



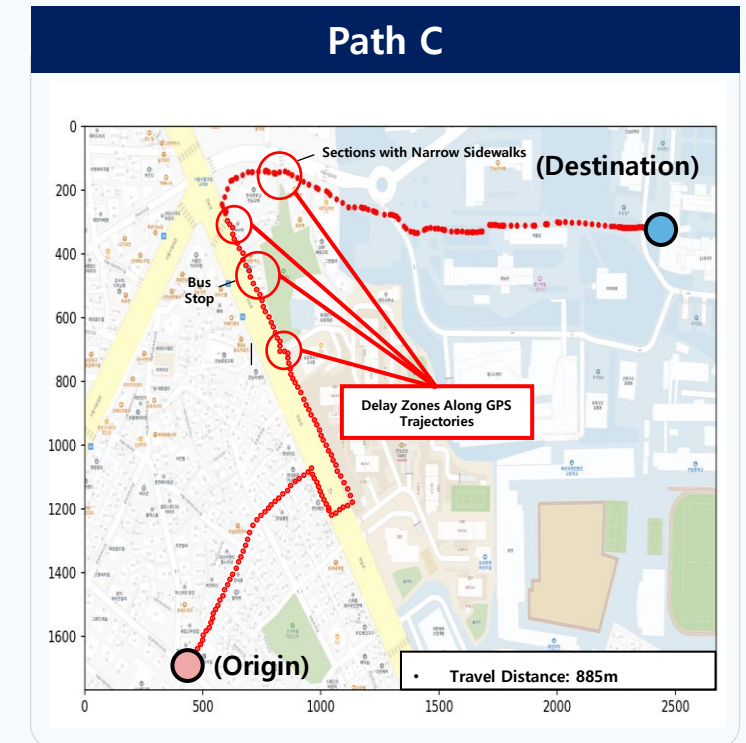
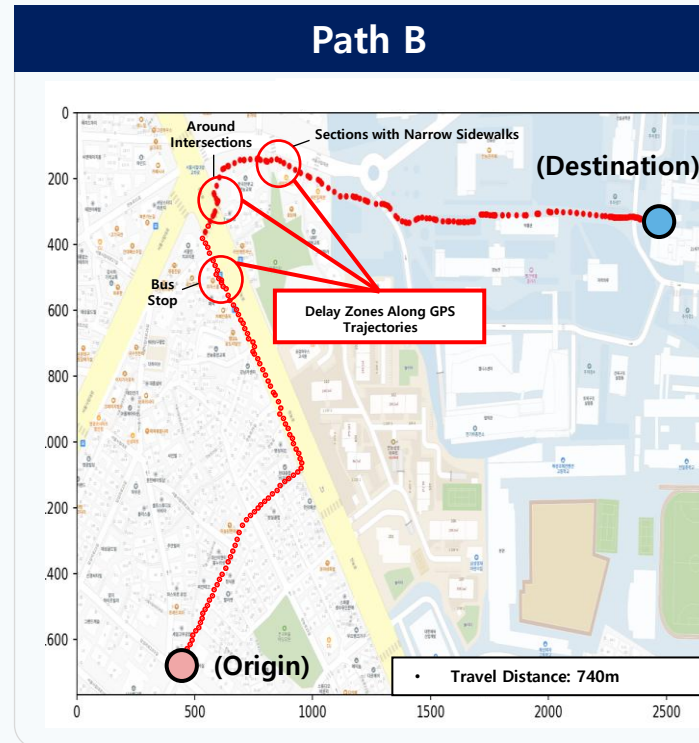
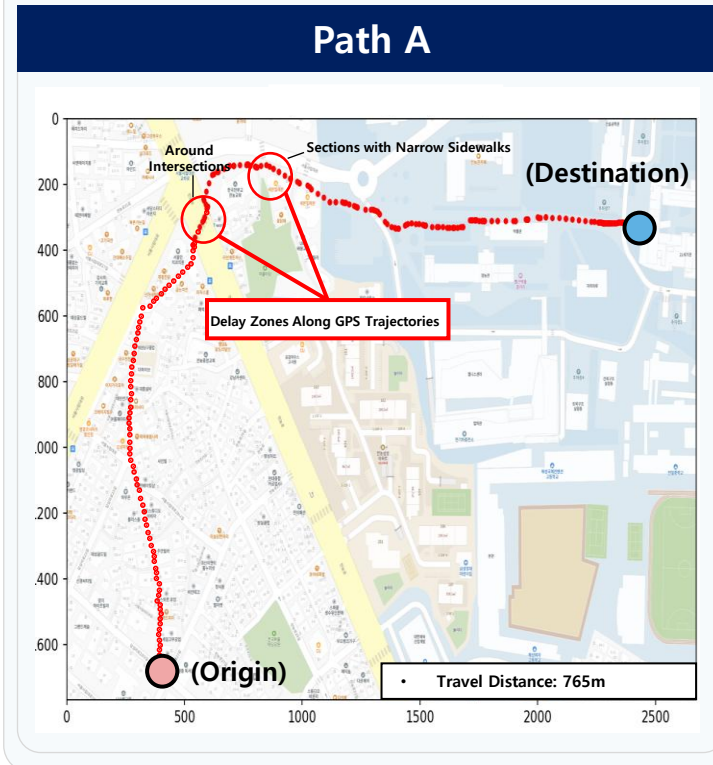
Bus Stop – Robot View



☑ GPS Trajectories and Observed Delay Zones by Path

- GPS trajectories were combined with field videos and observation records to identify recurrent slowing and temporary stops.
- Recurrent delays were observed near intersections, bus stops, and narrow sidewalk sections.
- Path C included the greatest number of observed delay zones.

Observed Delay Zones along GPS Trajectories by Path

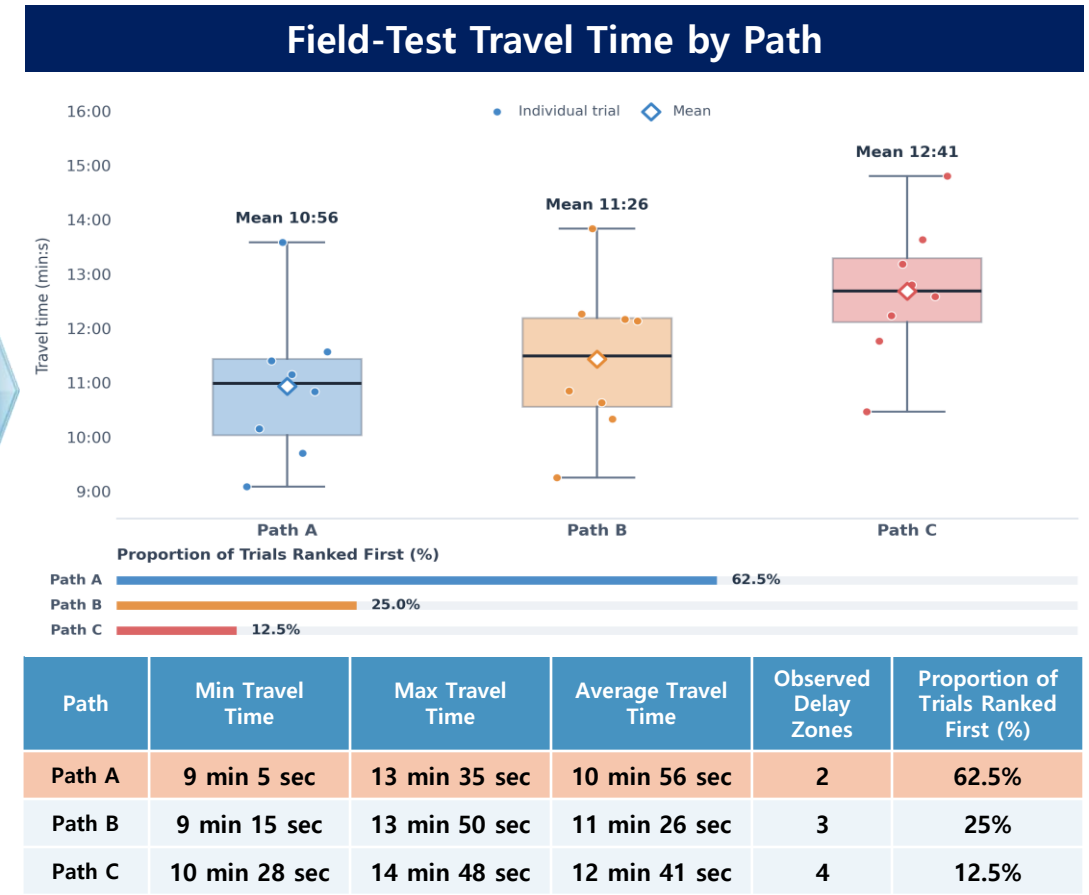


✓ Field-Test Travel Time Results by Path

- Path A recorded the shortest average and minimum travel times and ranked first in five of the eight field-test trials. Path C recorded the longest average travel time.
- Path A ranked first in 62.5% of the field-test trials, more often than the other paths.

Selected Field-Test Travel Times by Trial							
Trial	Path	Travel Time	Rank	Trial	Path	Travel Time	Rank
Trial 1	A	9min 5sec	1	Trial 4	A	13min 35sec	2
	B	9min 15sec	2		B	13min 50sec	3
	C	10min 28sec	3		C	13min 11sec	1
Trial 2	A	10min 9sec	1	Trial 5	A	11min 9sec	2
	B	10min 51sec	2		B	10min 38sec	1
	C	11min 46sec	3		C	12min 48sec	3
Trial 3	A	11min 24sec	1	Trial 6	A	9min 42sec	1
	B	12min 16sec	3		B	10min 20sec	2
	C	12min 14sec	2		C	13min 38sec	3

⋮



✓ Integrated Comparison of Path-Assessment Results

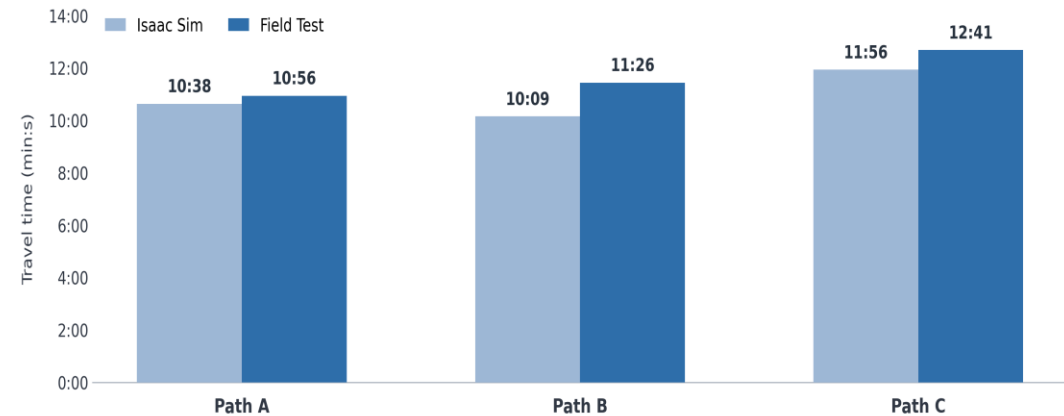
- UNA indicated the lowest network-based pedestrian movement potential for Path A and the highest for Path C.
- Isaac Sim showed the travel-time order B–A–C, whereas the field test showed A–B–C.
- Path C combined the highest network-based pedestrian movement potential with the longest travel time in both simulation and field testing.
- The three methods provide complementary information for evaluating candidate AMR paths.

UNA and Travel-Time Ranking

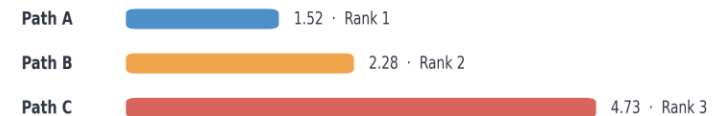
Method	A	B	C
UNA	1.52	2.28	4.73

Method	1 st	2 nd	3 rd
Isaac Sim	B (10min 09sec)	A (10min 38sec)	C(11min 56sec)
Field Test	A(10min 56sec)	B(11min 26sec)	C(12min 41sec)

Isaac Sim & Field-Test Travel Time by Path



UNA-based route assessment



05 Conclusion

✓ Conclusion

- AMR path performance varied across simulation and field conditions, reflecting differences in path distance, geometry, and real-world interactions.
- UNA, simulation, and field testing provide complementary information for AMR Path planning.

✓ Contribution

- Proposed an integrated framework linking urban network analysis, robot simulation, and field testing.
- Demonstrated a path-selection approach that considers both robot travel efficiency and network-based pedestrian movement potential.
- Provided field-based evidence on how pedestrian interactions and urban infrastructure conditions are associated with AMR travel performance.

✓ Future Research

- Expand the framework to multiple urban areas with different network structures and pedestrian conditions.
- Incorporate additional path-specific factors, including pedestrian crossings, roadside activities, and interactions with bicycles and personal mobility devices.
- Develop more reliable AMR path-planning strategies that improve operational efficiency while minimizing impacts on pedestrian comfort and safety.



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Thank you.

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